

HiLoTEL: Virtual Reality Robot Interface-Based Human-in-the-Loop Task Execution and Learning in the Physical World Through Its Digital Twin.

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ABSTRACT

HiLoTEL is a flexible virtual-reality framework for executing and learning tasks in virtual and physical environments. It enables human experts to collaborate with learning agents and intervene when necessary through human-in-the-loop imitation learning. HiLoTEL reduces the need to carry out repetitive tasks, providing the user with an intuitive supervision interface. The system is tested on Pick-and-Place task, considering both teleoperated and passthrough interaction modalities. The results show that HiLoTEL improves success rates while maintaining human-level completion time and providing users with 71% hands free supervision time, thus enabling effective human-robot collaboration.

Index Terms: Human-centered computing—Human computer interaction (HCI); Human-in-the-loop learning—Imitation learning; Human-robot collaboration

1 INTRODUCTION

Virtual reality (VR) technologies enable users to interact with simulated environments through immersive sensory feedback. VR finds applications in diverse domains, including education, engineering, and healthcare [1, 2]. However, one of the most challenging and promising applications of VR is in the teleoperation of robots and digital twins, allowing users to remotely control and manipulate robots in inaccessible or hazardous physical environments [1].

However, standard teleoperation approaches lack a sufficient degree of immersion and scene awareness, which could reduce the overall task performance [5]. VR offers several advantages for robot teleoperation, including immersive visualization, freedom of user viewpoint selection, and fluid interaction through natural action interfaces [3]. However, there are also challenges, such as communication latency, bandwidth limitations, synchronization issues and the impacts of prolonged usage of VR headsets. These limitations can be critical in teleoperation contexts, resulting in unresponsive and unstable systems that pose safety risks. Therefore, increasing the degree of autonomy in teleoperated systems through the adoption of execution and learning techniques involving the human operator may be beneficial.

In this paper, we introduce HiLoTEL, a novel system that uses VR to control and train robots in digital and real environments, exploiting either passthrough or teleoperation mode. HiLoTEL propose a human-in-the-loop imitation learning framework, for learning task execution policies from human experts, alleviating the burden of repetitive tasks while offering intervention capabilities and 3D interface feedback for supervision. The results obtained from a realistic Pick-and-Place experiment demonstrate that HiLoTEL improves task success rates and maintains human-level completion times. Additionally, it introduces flexibility, allowing users to take breaks from extended VR headset use.

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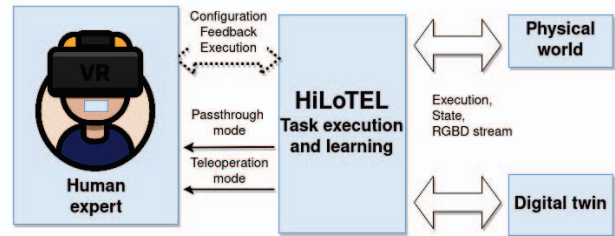


Figure 1: System architecture and data flow of HiLoTEL

2 SYSTEM DESCRIPTION

HiLoTEL consists of four main components: the VR-based robot interface, the task execution and learning module, the digital twin and the physical world. Figure 1 shows the overview of the system architecture and the data flow among the components.

The VR-based robot interface is the component that allows the human expert to control and interact with the robot in the physical environment through VR. The interface has two modes: teleoperation and passthrough. In teleoperation mode, the human expert can view and manipulate the digital twin in the VR scene, which is synchronized with the physical environment. The interface provides robot state, visual feedback, task status, and user interface components for system setup. In passthrough mode, the human expert can view and manipulate the real robot in the physical environment directly through the VR headset, which has a camera that captures the real-world scene.

The digital twin represents the virtual model of the physical environment, including the robot, in the VR scene. It is updated in real-time with sensor data from the physical environment over the network, incorporating the estimated states of robots and objects. The digital twin provides input data for the learning agent, including the robot's state, environment, and human expert actions.

The task execution and learning module is a key component in HiLoTEL, utilizing human-in-the-loop machine learning [6] to learn the task execution policy from a human expert. Through data aggregation techniques [4], this module collects data from the environment during expert interaction. The learning agent employs imitation learning, iteratively training a policy using supervised learning on expert-derived observation-action pairs. It also allows the trained agent to operate autonomously or in collaborative mode with the VR-based robot interface.

3 EXPERIMENTAL SETUP AND RESULTS

3.1 Experimental Setup

The experimental setup consists of a 7 DoF UFactory xArm7 robotic manipulator with a two-finger gripper, an Oculus Quest 2 VR headset, Depth Camera D455 and a workstation, as shown in Figure 2. We consider a representative Pick-and-Place task, requiring the robot to move a white bottle from the labeled pick position to the place position in less than 60 seconds, avoiding collisions with distractor objects. The human operator controlled the robot via Oculus Quest

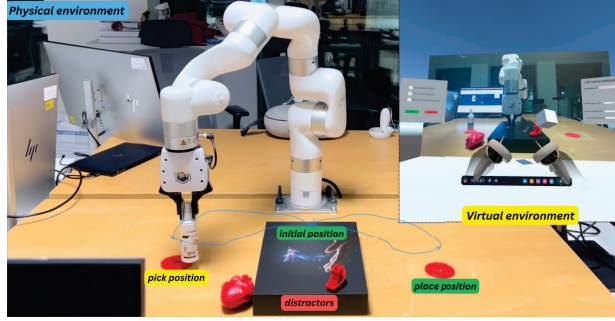


Figure 2: Picture of the experimental validation

2 in teleoperation and passthrough modes. As the human operator executed the task, the imitation learning agent collected data for task execution policy learning.

We assess HiLoTEL across four scenarios, evaluating task completion time, success rate, and hands-free time (e.g. time the human operator spent in supervision state). In the first two scenarios, the user performs the task using teleoperation and passthrough modes, as previously described. In the third scenario, the robot autonomously executes the task without human intervention, termed as the ‘pure-agent.’ Conversely, in the last scenario, the autonomous robot execution is supervised by the human operator, allowing intervention when necessary, such as in the event of a failed grasping, referred to as ‘HiLoTEL.’ For each scenario the user performed 15 trials. Task completion time measures how long it takes for either the human operator or the autonomous robot to complete the light blue ideal pick-and-place loop (Figure 2): initial position, pick, place, and return. Success rate is the percentage of successful tasks among trials, defined by accurate picking within the radius of the red circle on the left, precise placing within the radius of the red circle on the right, and retreating back to the initial position—all accomplished within 60 seconds and without collisions.

3.2 Results and Discussions

The results reported in Table 1 reveals that HiLoTEL had the highest success rate together with Human Teleoperation (93.33%), but with an higher hands-free time (71%), while maintaining a comparable completion time (24.58 vs 24.08 seconds). This suggests that HiLoTEL is more effective and efficient than the other scenarios, as it can achieve the best tradeoff between autonomy and collaboration. Human Teleoperation and Passthrough have a zero hands-free time, meaning that the human operator had to constantly supervise and control the agent, which could lead to fatigue and stress. Passthrough reduces the task completion time by 19.44(%) compared to Teleoperation, despite a 16.66(%) decrease in success rate. HiLoTEL enhances the task success rate by 40% compared to the pure-agent method, which had the lowest success rate (66.67%) and the shortest completion time (22.61 seconds), indicating that it is less reliable and accurate than the human-controlled scenarios.

Out of 15 trials, the Pure-agent completed 10, HiLoTEL completed 14 (10 autonomously, 4 collaboratively), while human completed 14 in Teleoperation and 12 in Passthrough. Passthrough failures are mainly robot collisions with distractors, often due to real-time rendering lag and accuracy issues. Quick passthrough completion is due to minimal communication overload compared to teleoperation. Pure-agent failures involve dropping objects, getting lost, and colliding with distractors—challenges related to the agent’s difficulty in generalizing to unseen states. HiLoTEL’s failure is from an irreversible agent action resulting in dropping an object, while human failure is attributed to a lack of attention.

Scenario	Completion Time (s)	Success Rate (%)	Hands-Free Time (%)
Human Passthrough	20.16 ± 4.02	80	0
Human Teleoperation	24.08 ± 2.77	93.33	0
Pure-Agent	22.61 ± 4.39	66.67	100
HiLoTEL	24.58 ± 2.86	93.33	71

Table 1: Experimental Results

4 CONCLUSION AND FUTURE WORK

In this paper, we introduce HiLoTEL a novel framework that employs VR to control and teach robots in both real and digital environments. It facilitates collaboration between human experts and learning agents, allowing interventions when necessary. The system also provides supervision feedback for autonomously running agents through VR. We showcase the capabilities of HiLoTEL using a Pick-and-Place task in both passthrough and teleoperation modalities, considering scenarios such as pure human, pure autonomous agent and collaborative human-agent interactions. The experimental results demonstrate the effectiveness of HiLoTEL, as it outperforms other considered approaches in terms of task success rates while maintaining completion times comparable to human performance. Additionally, HiLoTEL offers users a significant hands-free supervision time. This underscores the system’s effective VR-based human-robot collaboration.

Currently, HiLoTEL presents some limitations related to the absence of a mechanism for intervening in irreversible agent actions, reliance on physical world models, and dependency on network and sensor data. We plan to resolve these limitations in current and future research activities, together with extending the experimental validation by considering an extended user-test with an higher number of users and tasks.

HiLoTEL is a novel and flexible framework for learning and executing tasks in virtual and physical environments. We hope that HiLoTEL can contribute to advancing human-robot collaboration by providing an efficient and adaptable system for task execution in virtual and physical environments.

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